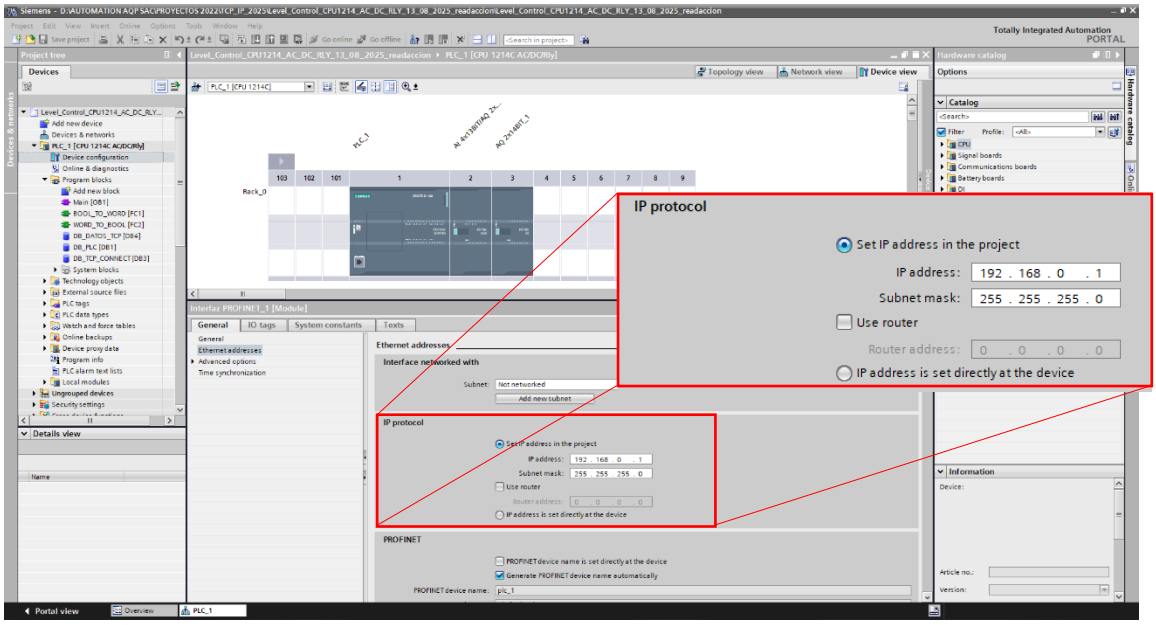
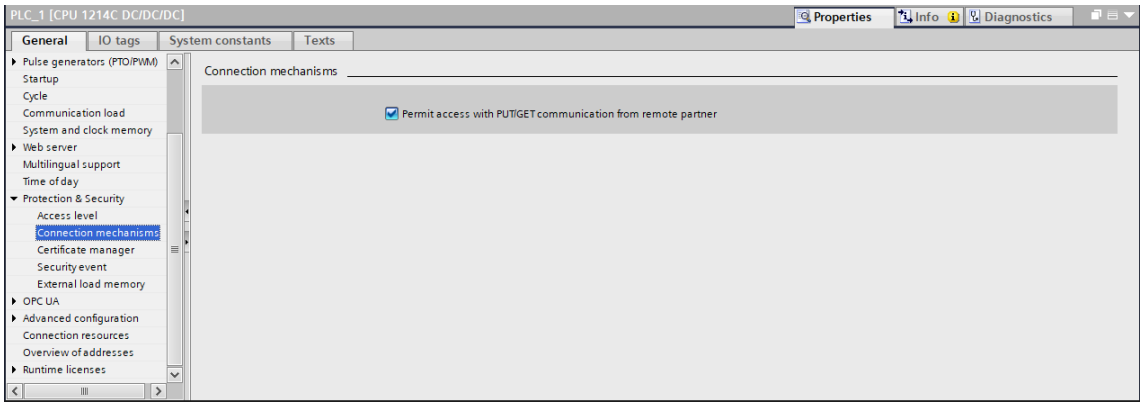
PLC Configuration

To allow the Python script to communicate with the S7-1200 via Modbus TCP:

1. Open **TIA Portal** and go to **Device Configuration**.
2. In the PLC Properties, ensure **Ethernet addresses** are correctly set.
   Example: IP address: 192.168.0.1 / Subnet mask: 255.255.255.0

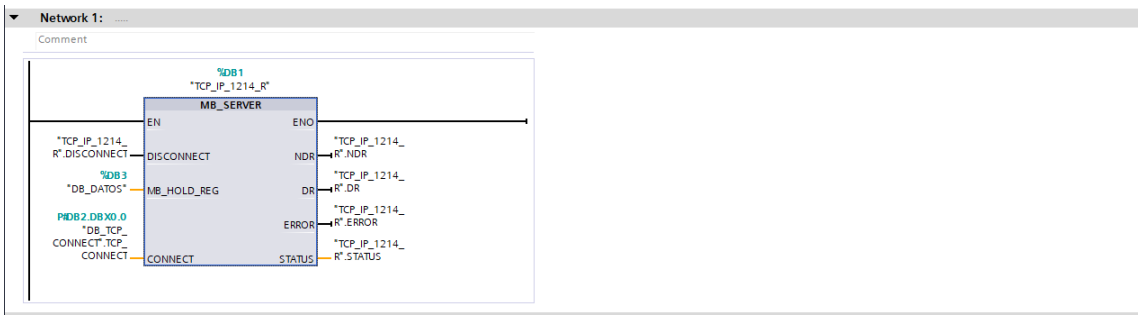


3. Under "Protection & Security," enable **"Permit access with PUT/GET communication from remote partner" to Firmware version V4.X**.



4. Configuration and the Modbus server block (MB_SERVER) to the PLC to Firmware version V2.X.

5. Configuration and the Modbus server block (MB_SERVER) to the PLC to Firmware version V4.X.



DB_TCP_CONNECT

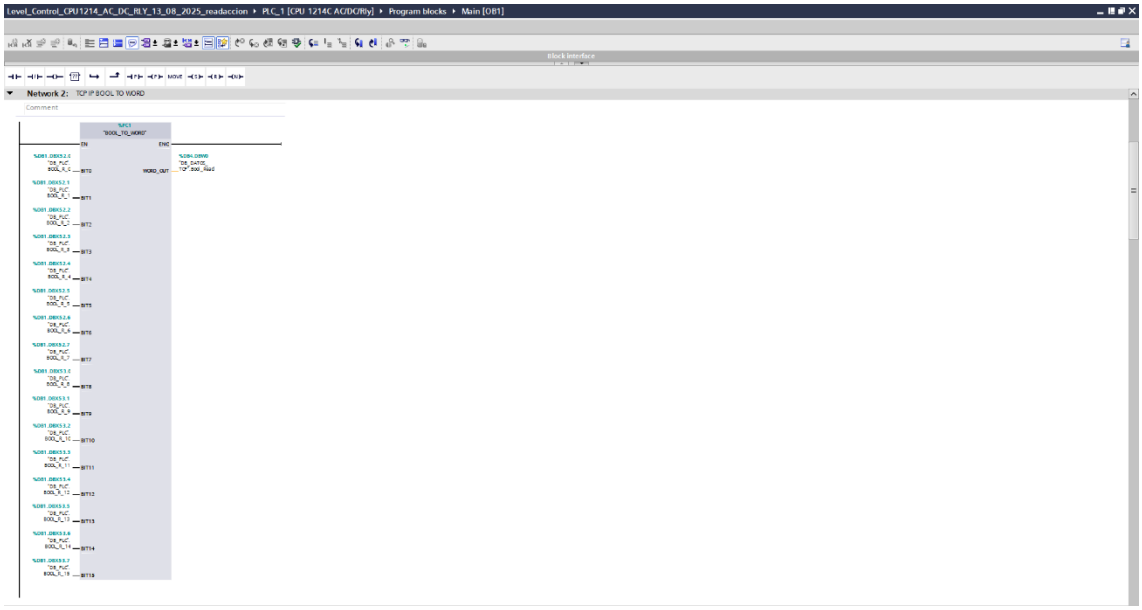
	Name	Data type	Offset	Start value	Retain	Accessible f...	Writa...	Visible in ...	Setpoint	Comment
1	Static									
2	TCP_CONNECT	TCON_IP_v4	0.0			✓	✓	✓		
3	InterfaceId	HW_ANY	0.0	64		✓	✓	✓		HW-identifier of IE-interface submodule
4	ID	CONN_OUC	2.0	1		✓	✓	✓		connection reference / identifier
5	ConnectionType	Byte	4.0	11		✓	✓	✓		type of connection: 11=TCP/IP, 19=UDP (17=TCP/IP)
6	ActiveEstablished	Bool	5.0	false		✓	✓	✓		active/passive connection establishment
7	RemoteAddress	IP_V4	6.0			✓	✓	✓		remote IP address (IPv4)
8	ADDR	Array[1..4] of Byte	6.0			✓	✓	✓		IPv4 address
9	ADDR[1]	Byte	6.0	192		✓	✓	✓		IPv4 address
10	ADDR[2]	Byte	7.0	168		✓	✓	✓		IPv4 address
11	ADDR[3]	Byte	8.0	0		✓	✓	✓		IPv4 address
12	ADDR[4]	Byte	9.0	60		✓	✓	✓		IPv4 address
13	RemotePort	UInt	10.0	0		✓	✓	✓		remote UDP/TCP port number
14	LocalPort	UInt	12.0	0		✓	✓	✓		local UDP/TCP port number

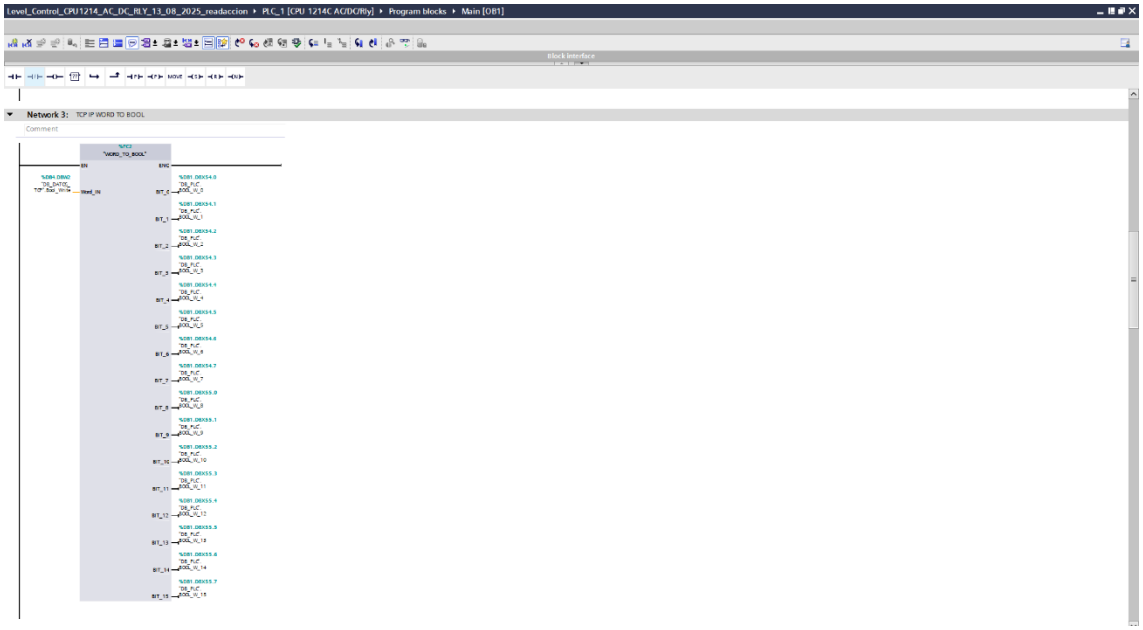
6. Data Block (DB) to Global Modbus Register Mapping

DB_DATOS_TCP

	Name	Data type	Offset	Start value	Retain	Accessible f...	Writa...	Visible in ...	Setpoint	Comment
1	Static									
2	Bool_Read	Word	0.0	16#0		✓	✓	✓		
3	Bool_Write	Word	2.0	16#0		✓	✓	✓		
4	Real_0	Real	4.0	0.0		✓	✓	✓		
5	Real_1	Real	8.0	0.0		✓	✓	✓		
6	Real_2	Real	12.0	0.0		✓	✓	✓		
7	Real_3	Real	16.0	0.0		✓	✓	✓		
8	Real_4	Real	20.0	0.0		✓	✓	✓		
9	Real_5	Real	24.0	0.0		✓	✓	✓		
10	Real_6	Real	28.0	0.0		✓	✓	✓		
11	Real_7	Real	32.0	0.0		✓	✓	✓		
12	Real_8	Real	36.0	0.0		✓	✓	✓		
13	Real_9	Real	40.0	0.0		✓	✓	✓		
14	Real_10	Real	44.0	0.0		✓	✓	✓		
15	Real_11	Real	48.0	0.0		✓	✓	✓		
16	Real_12	Real	52.0	0.0		✓	✓	✓		
17	Real_13	Real	56.0	0.0		✓	✓	✓		
18	Real_14	Real	60.0	0.0		✓	✓	✓		
19	Real_15	Real	64.0	0.0		✓	✓	✓		

7. Modbus Data Conversion Functions (FC), Custom Function Blocks designed for Bit-to-Word packing and Word-to-Bit unpacking. These functions facilitate the mapping of Boolean status indicators into 16-bit Modbus registers


Level_Control_CPU1214_AC_DC_RLY_13_08_2025_readaccion ▸ PLC_1 [CPU 1214C AC/DC/Rly] ▸ Program blocks ▸ Main [OB1]
Block interface
Network 2: TCP IP BOOL TO WORD
Comment
%FC1
"BOOL_TO_WORD"
EN
ENO
%DB1.DBX52.0
"DB_PLC_
BOOL_0_0"
BIT0
%DB1.DBX52.1
"DB_PLC_
BOOL_0_1"
BIT1
%DB1.DBX52.2
"DB_PLC_
BOOL_0_2"
BIT2
%DB1.DBX52.3
"DB_PLC_
BOOL_0_3"
BIT3
%DB1.DBX52.4
"DB_PLC_
BOOL_0_4"
BIT4
%DB1.DBX52.5
"DB_PLC_
BOOL_0_5"
BIT5
%DB1.DBX52.6
"DB_PLC_
BOOL_0_6"
BIT6
%DB1.DBX52.7
"DB_PLC_
BOOL_0_7"
BIT7
%DB1.DBX53.0
"DB_PLC_
BOOL_0_8"
BIT8
%DB1.DBX53.1
"DB_PLC_
BOOL_0_9"
BIT9
%DB1.DBX53.2
"DB_PLC_
BOOL_0_10"
BIT10
%DB1.DBX53.3
"DB_PLC_
BOOL_0_11"
BIT11
%DB1.DBX53.4
"DB_PLC_
BOOL_0_12"
BIT12
%DB1.DBX53.5
"DB_PLC_
BOOL_0_13"
BIT13
%DB1.DBX53.6
"DB_PLC_
BOOL_0_14"
BIT14
%DB1.DBX53.7
"DB_PLC_
BOOL_0_15"
BIT15
WORD_OUT
%DB4.DBW0
"DB_DATOS_
TCP".BOOL_WORD


Level_Control_CPU1214_AC_DC_RLY_13_08_2025_readaccion ▸ PLC_1 [CPU 1214C AC/DC/Rly] ▸ Program blocks ▸ Main [OB1]
Block interface
Network 3: TCP IP WORD TO BOOL
Comment
%FC2
"WORD_TO_BOOL"
EN
ENO
%DB4.DBW2
"DB_DATOS_
TCP".BOOL_WORD
Word_IN
BIT_0
%DB1.DBX54.0
"DB_PLC_
BOOL_IX_0"
BIT_1
%DB1.DBX54.1
"DB_PLC_
BOOL_IX_1"
BIT_2
%DB1.DBX54.2
"DB_PLC_
BOOL_IX_2"
BIT_3
%DB1.DBX54.3
"DB_PLC_
BOOL_IX_3"
BIT_4
%DB1.DBX54.4
"DB_PLC_
BOOL_IX_4"
BIT_5
%DB1.DBX54.5
"DB_PLC_
BOOL_IX_5"
BIT_6
%DB1.DBX54.6
"DB_PLC_
BOOL_IX_6"
BIT_7
%DB1.DBX54.7
"DB_PLC_
BOOL_IX_7"
BIT_8
%DB1.DBX55.0
"DB_PLC_
BOOL_IX_8"
BIT_9
%DB1.DBX55.1
"DB_PLC_
BOOL_IX_9"
BIT_10
%DB1.DBX55.2
"DB_PLC_
BOOL_IX_10"
BIT_11
%DB1.DBX55.3
"DB_PLC_
BOOL_IX_11"
BIT_12
%DB1.DBX55.4
"DB_PLC_
BOOL_IX_12"
BIT_13
%DB1.DBX55.5
"DB_PLC_
BOOL_IX_13"
BIT_14
%DB1.DBX55.6
"DB_PLC_
BOOL_IX_14"
BIT_15
%DB1.DBX55.7
"DB_PLC_
BOOL_IX_15"